

GPU-accelerated reconstruction of tiled spectral-domain OCT with removal of specular film reflections

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Abstract

Tiled spectral-domain optical coherence tomography (OCT) can image centimetre-scale tissue at micrometre resolution, but reconstruction is slow and samples mounted between transparent films show bright specular film reflections that mask the tissue surface. We present an open, GPU-accelerated reconstruction pipeline that reproduces the established myOCT tiled-scan reconstruction to within one output quantisation step while reducing the time for a 956-GB, 864-tile volume from an estimated 25.5 h to 18.4 min, limited only by storage bandwidth. We further introduce a per-A-line method that removes reflections of fluorinated ethylene propylene (FEP) films in the complex depth profile: each reflection is projected onto a rank-3 point-spread-function subspace learnt from the scan itself and shrunk to the local background level, so no dark band is left behind. On tissue, the 99th percentile of the surface en-face plane drops by 9.8 dB, deep tissue is unchanged, and the removed signal is exported for inspection. With the reflections gone, an automatically segmented tissue slab yields a tissue-only en-face projection: a single two-dimensional image of the section. The removal runs within the same GPU pass and adds 2% to the full-volume reconstruction time.

Introduction

Optical coherence tomography (OCT) measures depth-resolved backscattering by low-coherence interferometry^{1,2}. In spectral-domain (Fourier-domain) OCT the depth profile of every lateral position (an A-line) is recovered from one spectrometer readout, which gives large gains in speed and sensitivity over time-domain systems^{3–6}. To image specimens much larger than the field of view of a high-numerical-aperture objective, the sample is translated under the scan head and a mosaic of tiles is acquired at several focal depths; the tiles are then corrected for field curvature, weighted by their distance to focus and stitched into one volume. The open-source MATLAB toolbox myOCT⁷ implements this tiled reconstruction for Thorlabs and Wasatch spectral-domain systems, and it is used to produce virtual-histology volumes.

Two problems limit this workflow at scale. First, reconstruction is slow: for the dataset studied here (12×9 lateral tiles at 8 focal depths, 864 tiles, 432,000 B-scans, 956 GB of raw spectra) the reference implementation needs 20.4 s per output plane, about 25.5 h for the 4500 planes, and its default parallel mode could not run on a 14-GB workstation. Graphics processing units (GPUs) are well established for real-time spectral-domain OCT processing of single B-scans^{8–10}, but a tiled, multi-depth reconstruction is dominated by data movement and by the bookkeeping of thousands of tiles, and to be useful it must reproduce the existing reference output exactly.

Second, many samples are mounted between thin transparent films—here fluorinated ethylene propylene (FEP)—to keep them flat and hydrated. The film surfaces are flat, smooth interfaces and act as mirrors. Each produces a specular reflection that is 10–30 dB brighter than the tissue and that lies, by construction, exactly at the tissue surface. In the stitched volume it appears as a bright band at $z \approx 0$ in every B-scan and, because the specular return depends on the angle between beam and film, as round, tile-periodic blobs in en-face planes and in projections. Lateral averaging, notch filtering or low-rank background subtraction either fail to suppress it or remove the tissue at the surface with it.

Here we describe **octrecon**, a Python reconstruction for NumPy/Numba (CPU) and CuPy/CUDA (GPU) that re-expresses each stage of the reference reconstruction as precomputed linear operators and fused kernels and streams the raw data tile by tile. We then introduce a physically motivated removal of film reflections that operates on the complex A-line before the magnitude is taken: the reflection is a single, shifted copy of the system point-spread function (PSF), whose complex depth profile spans a three-dimensional subspace, whereas tissue speckle spreads over all dimensions. Projecting onto that subspace and shrinking the projection down to the local background level removes the film without creating a dark band and leaves every sample outside a 17-sample window bit-identical. Finally, we use the cleaned volume to segment the tissue and export a tissue-only en-face projection, a single two-dimensional image of the section to which neither air, film nor background contribute.

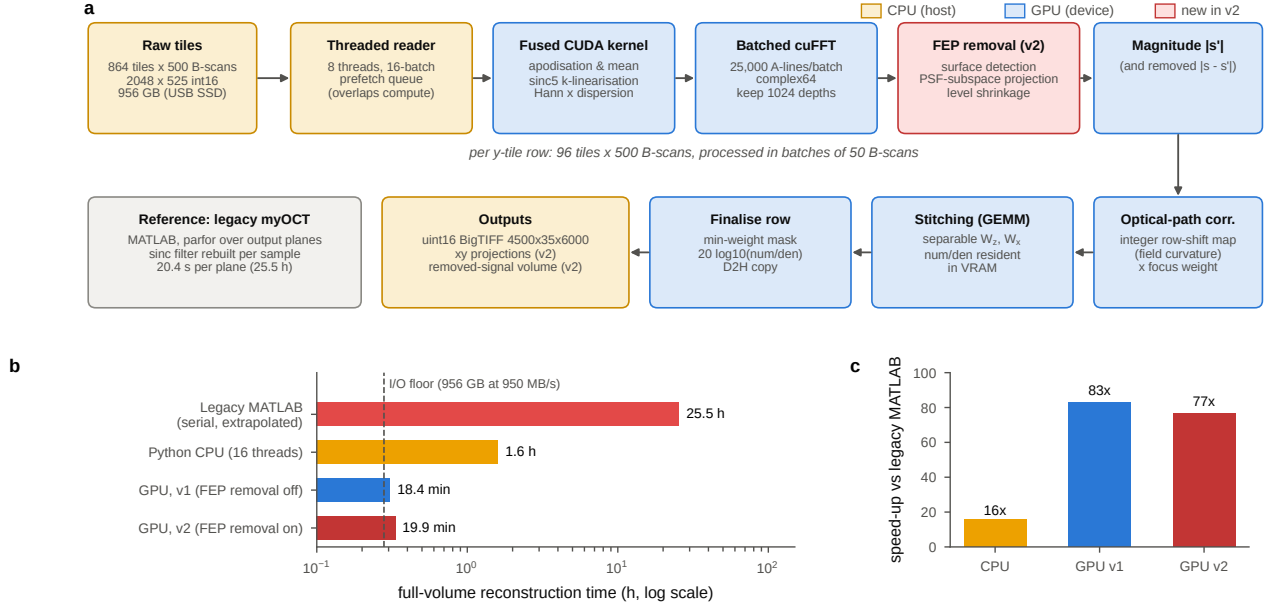


Fig. 1. Reconstruction pipeline and acceleration. **a**, Data flow for one row of tiles (96 tiles $\times$ 500 B-scans). Orange, host (CPU); blue, GPU; red, FEP-film reflection removal added in this work. The FEP step operates on complex depth profiles before the magnitude; the stitched magnitude of the removed component is exported as a separate volume. The grey box summarises the reference implementation (myOCT, MATLAB). **b**, Full-volume reconstruction time (log scale) for the dataset (864 tiles, 956 GB raw, 4500 $\times$ 35 $\times$ 6000 output). Reference time is a measured plane (20.4 s) $\times$ 4500. CPU and GPU v1 values are measured end-to-end runs including the BigTIFF output. Dashed line, time to read 956 GB at 950 MB s $^{-1}$. **c**, Speed-up relative to the reference. GPU v2 values (FEP removal on) are filled in from the full-volume run (19.9 min).

Results

A streaming GPU reconstruction that reproduces the reference output

The pipeline (Fig. 1a) follows the reference reconstruction step by step: subtraction of the apodisation spectrum, resampling to linear wavenumber with a five-lobe sinc kernel (sinc5), a Hann window with a quadratic dispersion phase, an inverse FFT, optical-path (field-curvature) correction, a Gaussian focus weight and a weighted-mean stitching into the output grid (Methods). Every operator that does not depend on the data—the sinc weights, the complex window, the curvature shift map and the interpolation matrices of the stitching—is computed once and kept on the device. The sinc5 resampling, which takes 77 % of the reference run time because the filter is rebuilt for each of the 2048 output samples of each B-scan, becomes a banded operator with at most 14 taps per sample and is fused with the apodisation and window into a single CUDA kernel; this alone made the spectral stage 6.2 times faster than an unfused GPU implementation. The 2D interpolations of each B-scan onto the full output plane become two small dense matrices per tile position, applied as batched matrix products to accumulators that stay in GPU memory for a whole row of tiles. Raw files are read tile-major by eight threads with 16 batches in flight, so disk reads overlap with computation.

On a laptop workstation (RTX 3080 Ti Laptop GPU, Ryzen 9 6900HX, 14 GB RAM, USB SSD) the full volume (4500 $\times$ 35 $\times$ 6000 output voxels at 2 μ m) was reconstructed in 18.4 min end to end, 83 times faster than the reference extrapolated from a measured plane, and 5.3 times faster than the same code on all 16 CPU threads (1.6 h; Fig. 1b,c). The GPU waited for the disk 56 % of the time and read the raw data at an effective 877 MB s $^{-1}$, close to the drive limit: the reconstruction is input/output bound and the I/O floor for 956 GB is $\approx$ 17 min. Numerically, each stage agrees with the unmodified reference code to floating-point round-off; a stitched plane agrees to 9.3×10^{-5} dB in single precision, an order of magnitude below the output quantisation step (1.02×10^{-3} dB). Over all 945,000,000 voxels the GPU output lies within one quantisation step (1 LSB) of the existing reference volume, with identical masks of missing data (Extended Data Fig. 1); with the reference’s two-step quantisation enabled it reproduces the reference file bit-exactly, which also identified the dispersion coefficient used for it ($\beta = 8.949 \times 10^7$ nm 2 rad $^{-1}$).

Specular reflections of the FEP film

The tissue was mounted between FEP films. In stitched B-scans the film/tissue interface appears as a bright band at $z \approx 0$ (Fig. 2a). In en-face planes near the surface the film produces bright, round blobs whose centre repeats with the tile period (Fig. 2b,c; most visible in the tissue-free tile at $x \approx +5.5$ mm): the film is specular,

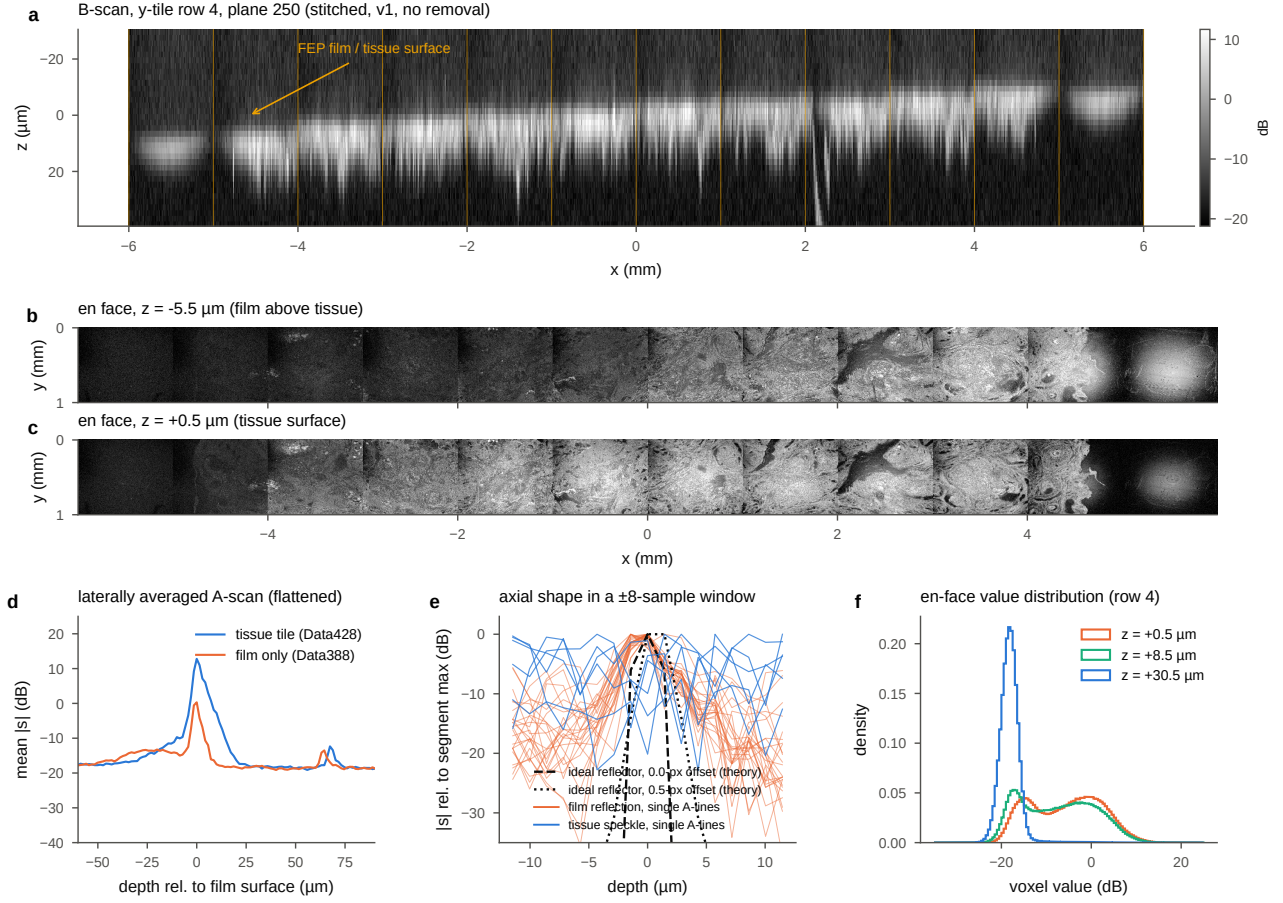


Fig. 2. Specular reflections of the FEP film. **a**, Stitched B-scan (row of tiles 4, plane 250; reconstruction without removal); vertical lines mark tile borders; vertical scale exaggerated. **b,c**, En-face planes of the same row of tiles (12×1 mm, y stretched $\approx 1.3\times$) at $z = -5.5 \mu\text{m}$ (film above the tissue) and $z = +0.5 \mu\text{m}$ (tissue surface). Note the round, tile-periodic blobs (clearest in the tissue-free tile on the right) and the tile-periodic brightness modulation. **d**, Laterally averaged, curvature-flattened magnitude of one B-scan of a tile with tissue (Data428) and of a tile with film only (Data388); 0 is the film surface, 1 pixel = $1.434 \mu\text{m}$ in tissue. **e**, Magnitude of 25 single A-lines of the film reflection (orange) and of six tissue A-lines (blue) in a ± 8 -sample window, each normalised to its maximum, with the theoretical response of an ideal reflector through the spectral window at integer and half-pixel position (black). **f**, Distribution of voxel values in en-face planes at the surface and in tissue (row of tiles 4).

so its return is strongest where beam and film are normal to each other, i.e. at the same position within every tile. The laterally averaged A-scan of a tissue tile shows a sharp peak 12.8 dB at the surface compared with a background of -13.9 dB (Fig. 2d), and the surface en-face plane has a bimodal value distribution with a bright mode that is absent in deep tissue (Fig. 2f).

Single A-lines reveal what distinguishes the film from tissue (Fig. 2e). Every film reflection has the same axial shape—the system PSF, a few pixels wide—and differs only by amplitude, phase and a sub-pixel shift; tissue speckle has a random axial structure. Across the field, the film amplitude fluctuates by about $\pm 20\%$ from A-line to A-line and its phase is random from B-scan to B-scan, which is why subtracting a laterally averaged reflection (in x or y) suppressed the film by only 10–17 dB in our tests; low-rank (SVD) background removal in a B-scan window failed because tissue at the surface is locally low-rank too, and notching the band destroys the tissue surface.

Removal by PSF-subspace projection and level shrinkage

A specular reflector at depth m contributes $a e^{i\varphi} p(z - m)$ to the complex A-line, where p is the complex PSF formed by the spectral window and the residual dispersion. For sub-pixel positions m these profiles span, within a 17-sample window ($\pm R$, $R = 8$), a subspace of very low dimension: one component captures 93.6% and three components $>99.99\%$ of the energy of the theoretical family (Fig. 3a,c). Real reflections are not exactly theoretical, because of residual dispersion; we therefore learn the basis from the scan itself, from about 6,000 strong, phase-aligned film segments of 12 tiles spread over the grid (Methods). The learnt rank-3 basis (Fig. 3b) captures 96.8% of the energy of a film reflection, against 85.5% for the theoretical PSF (Fig. 3c,d; on the 2,082 segments of the two example tiles: 96.0% versus 84.3%).

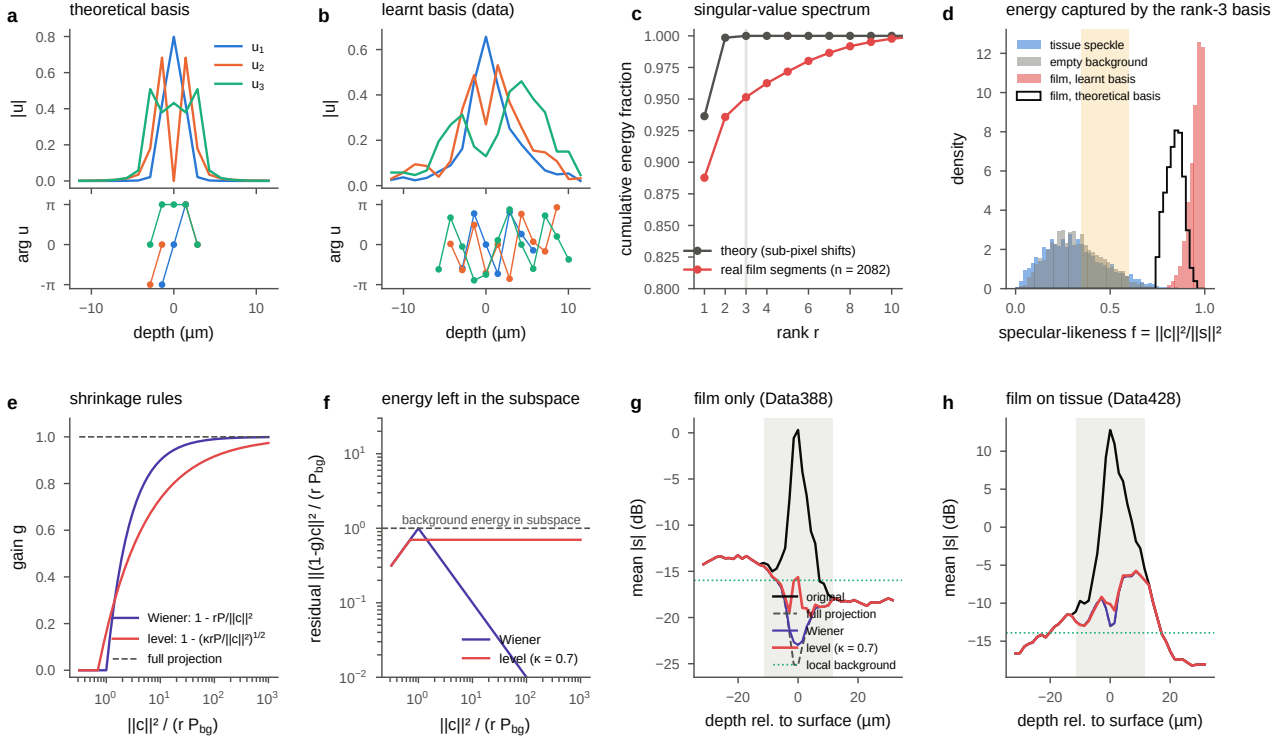


Fig. 3. Point-spread-function subspace and shrinkage rule. **a**, Rank-3 basis computed from the theoretical response of a reflector at 41 sub-pixel positions through the spectral window (magnitude, top; phase where $|u| > 0.1$, bottom). **b**, Rank-3 basis learnt from phase-aligned film segments of two tiles (Data388, Data428; 5 B-scans each). **c**, Cumulative energy fraction of the singular values for the theoretical family and for 2,082 real film segments. The grey line marks the rank used. **d**, Fraction of segment energy captured by the rank-3 basis for tissue speckle (Data428, 16–58 samples below the surface), empty background (Data388) and film segments (learnt and theoretical basis). Yellow, ramp of the soft weight w (0.35–0.60). **e**, Gain g versus the ratio of the projected energy to the expected background energy in the subspace, for the Wiener and the level rule ($\kappa = 0.7$). **f**, Energy left in the subspace after shrinkage, relative to the expected background energy; the level rule leaves κ of it for any film amplitude, the Wiener rule removes it as the film gets stronger. **g,h**, Laterally averaged, curvature-flattened profiles of one B-scan before and after removal for a film-only tile and a tissue tile. Grey band, $\pm R$ window; dotted, local background (10 rows on each side outside the window). Full projection and Wiener rule dig a trench below the background; the level rule does not.

For every A-line the method (i) locates each film surface, detected as a sharp peak of the laterally averaged, curvature-flattened profile and refined per A-line within ± 4 samples followed by a 15-A-line lateral median, (ii) projects the complex 17-sample segment $\mathbf{s}$ onto the basis $\mathbf{U}$, $\mathbf{c} = \mathbf{U}^H \mathbf{s}$, (iii) uses the fraction of energy captured, $f = \|\mathbf{c}\|^2 / \|\mathbf{s}\|^2$ (‘specular-likeness’), to decide whether a film is present, and (iv) subtracts the projection, $\mathbf{s}' = \mathbf{s} - w g \mathbf{U} \mathbf{c}$. The weight w ramps from 0 to 1 between $f = 0.35$ and 0.60 (after a lateral median); film segments lie almost entirely above the ramp, whereas tissue and empty background have median f of 0.30 and 0.31 (Fig. 3d; Extended Data Fig. 2). Speckle has f above the white-noise value $r/(2R+1) = 0.18$ because it is axially band-limited by the same PSF; it is therefore excluded by the surface detection and by the ramp together, and processing is confined to detected surfaces.

The gain g is the key to avoiding artefacts. Tissue and noise also have energy along the PSF subspace, on average about $r P_{bg}$ for a local background power P_{bg} per sample. Subtracting the whole projection ($g = 1$), or shrinking it with a Wiener-type rule $g = 1 - r P_{bg} / \|\mathbf{c}\|^2$, removes this share as well and leaves a dark trench where the film was (Fig. 3e–g): on a film-only tile the surface fell from 0.3 dB to –25.2 dB (full projection) or –23.0 dB (Wiener), well below the –16.0 dB background. We instead shrink the coefficient to the background level, $g = \max\left(0, 1 - \sqrt{\kappa r P_{bg} / \|\mathbf{c}\|^2}\right)$ with $\kappa = 0.7$, which leaves exactly $\kappa r P_{bg}$ of energy in the subspace whatever the film amplitude (Fig. 3f). The surface then lands at –15.6 dB, at the background level, and on a tissue tile the 12.8 dB peak is reduced to –10.1 dB next to a local background of –13.9 dB (Fig. 3h). Samples outside the window are not modified.

Validation on synthetic ground truth

We injected PSF reflections with smooth sub-pixel tilt, random phase per A-line and $\pm 20\%$ amplitude jitter, as measured on the film, into simulated speckle tissue (Extended Data Fig. 3). Without removal the error in

Table 1. En-face statistics before and after film-reflection removal (row of tiles 4, 500×6000 pixels per plane, values in dB as stored in the output volume).

z (μm)	region	median			99th percentile		
		before	after	Δ	before	after	Δ
-5.5	film above tissue	-10.7	-14.2	-3.5	7.9	-3.0	-10.9
+0.5	tissue surface	-5.1	-12.9	-7.8	10.2	0.4	-9.8
+4.5	tissue	-5.1	-12.3	-7.2	10.7	2.7	-8.0
+8.5	tissue	-7.4	-13.0	-5.7	9.5	3.7	-5.8
+14.5	tissue	-14.7	-15.9	-1.2	6.0	3.3	-2.7
+30.5	deep tissue	-18.2	-18.2	0.0	-9.4	-9.8	-0.4

the 17-sample window was +1.7, +11.7 and +21.7 dB relative to the tissue for peaks 10, 20 and 30 dB above the tissue. With the rank-3 basis and the level rule it was -7.8, -6.9 and -6.6 dB, i.e. independent of the film amplitude; a rank-1 basis failed for strong reflections (+10.6 dB at 30 dB) because it cannot follow sub-pixel shifts. Full-projection and Wiener rules gave errors about 2 dB lower in tissue (-8.8 and -8.8 dB at 20 dB) but, on an empty background, the level rule kept the surface at the background level (+0.4 dB) and on real data only the level rule avoided the trench (Fig. 3g). Tissue without a reflection was left unchanged (the weight is zero), and reflections at or below the tissue level are deliberately left in place because they are not separable from speckle. The same criteria are enforced as unit tests of the software.

Real tissue volumes

Applied to a full row of tiles (500 output planes, 12×1 mm), removal eliminated the bright surface band and the tile-periodic blobs while preserving tissue structure (Fig. 4). At the tissue surface ($z = +0.5 \mu\text{m}$) the median en-face value dropped from -5.1 to -12.9 dB and the 99th percentile from 10.2 to 0.4 dB; in the film plane above the tissue ($z = -5.5 \mu\text{m}$) the 99th percentile dropped from 7.9 to -3.0 dB (Table 1). The effect decays with depth as the film window leaves the output plane, and deep tissue ($z = +30.5 \mu\text{m}$) is unchanged (median -18.2 versus -18.2 dB). Laterally, the y-averaged surface plane loses its tile-periodic arcs and the blob in the tissue-free tile disappears (Fig. 4d). The difference images (Fig. 4a, right) show that what is removed is concentrated in the two to three planes nearest the film and follows the illumination pattern of the tile rather than tissue morphology.

Tissue-only en-face projection and the removed-signal volume

For many uses a single two-dimensional image of the section is wanted. Version 2 therefore exports a calibrated *tissue-only* en-face projection: the tissue is segmented automatically in the reconstructed volume (lateral smoothing of the amplitude over 30 μm , Otsu threshold, hole filling; Methods), each column is reduced to the slab between its upper and lower tissue surface, and only voxels inside this slab are averaged (mean amplitude P_{mean}) or maximised (P_{max}). Air, film, background and columns without tissue do not contribute and are left empty. In addition the pipeline writes a ‘removed-signal’ volume: the stitched magnitude of exactly the component that was subtracted, $|\mathbf{s} - \mathbf{s}'|$, using the same curvature correction and stitching weights as the image. The removed volume makes the operation auditable: any tissue structure in it would indicate over-removal.

Film removal is a prerequisite for the tissue-only projection. With the same segmentation, the film reflections in tiles without tissue pass the threshold, so the ‘tissue’ footprint covers 82 % of the field without removal (round, tile-periodic blobs around the section, Fig. 5a) but 62 % with removal, which is the section itself (Fig. 5b; threshold -12.1 dB, median slab thickness 22 μm). Inside the tissue the bright surface band otherwise dominates every depth average. For one row of tiles, removal lowered the median of P_{mean} from -10.2 to -14.5 dB and the tile-periodic modulation (Methods) from 7.5 to 4.5 dB (Extended Data Fig. 4). For the full volume (Fig. 5) the tile-periodic modulation of P_{mean} is 6.7 dB without and 4.8 dB with removal, the median difference between the two projections is 7.0 dB, and the median removed amplitude is -13.5 dB (99th percentile 0.5 dB). All three outputs and the removal itself can be switched off, in which case the pipeline produces output identical to version 1.

Runtime

Removal runs inside the same GPU pass, on the complex batch that is already on the device and only on the depth band that reaches the output; only small per-A-line arrays (peak rows, weights) are moved to the host for the lateral medians. On one row of tiles the GPU time rose from 106 s to 150–190 s (Fig. 6b). For the full

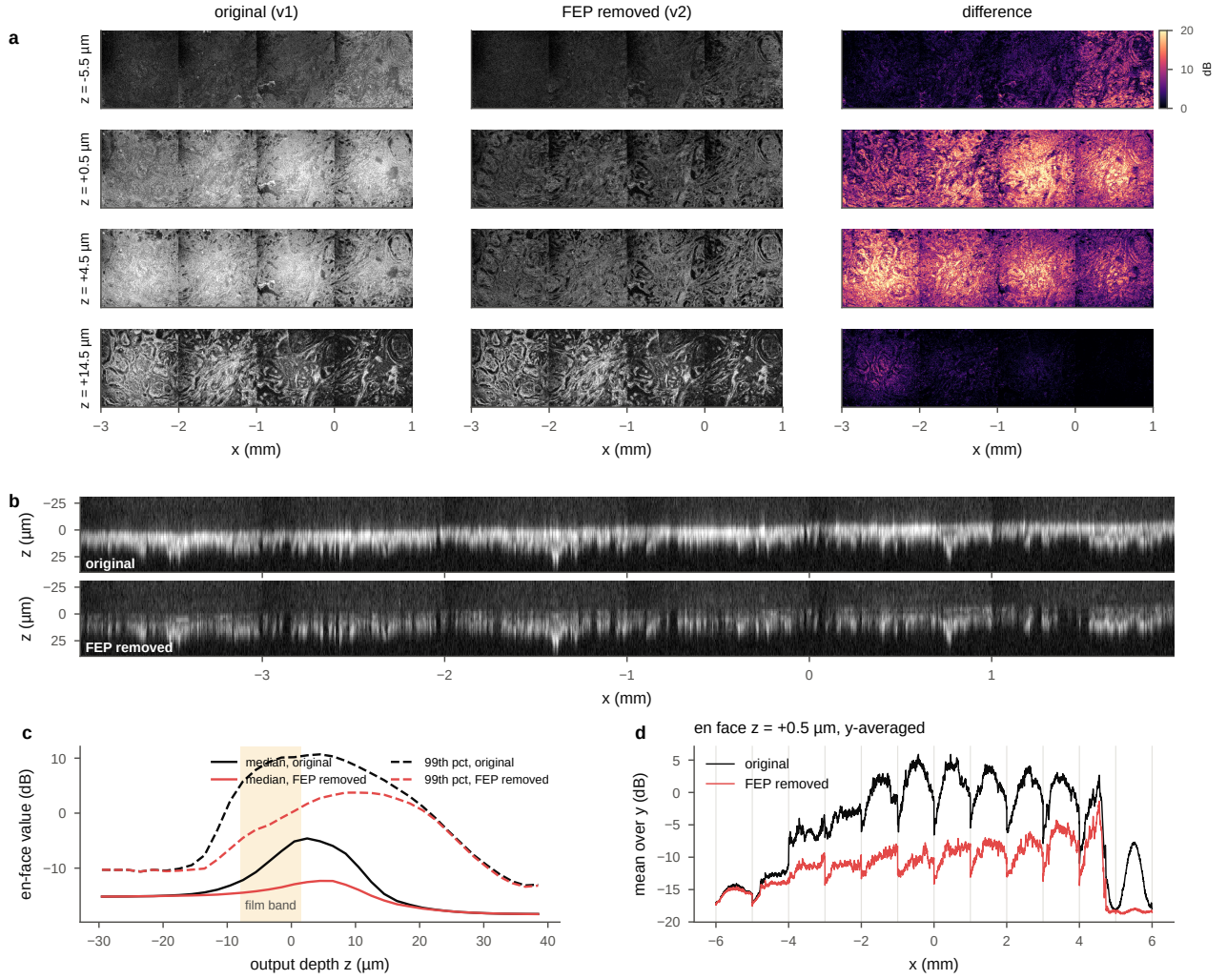


Fig. 4. Film-reflection removal on a stitched tissue volume. **a**, En-face planes at four depths of row of tiles 4 (4 mm of the 12-mm row shown) before (left) and after (middle) removal, displayed with the same grey scale (2nd–99.7th percentile of the original plane), and their difference in dB (right). **b**, B-scan (plane 250, 6 mm shown) before and after removal; vertical scale exaggerated. **c**, Median and 99th percentile of the en-face planes versus output depth. Yellow, depth range of the film reflection. **d**, Surface en-face plane averaged over y (amplitude average, in dB) across the whole row of tiles; grey lines mark tile borders.

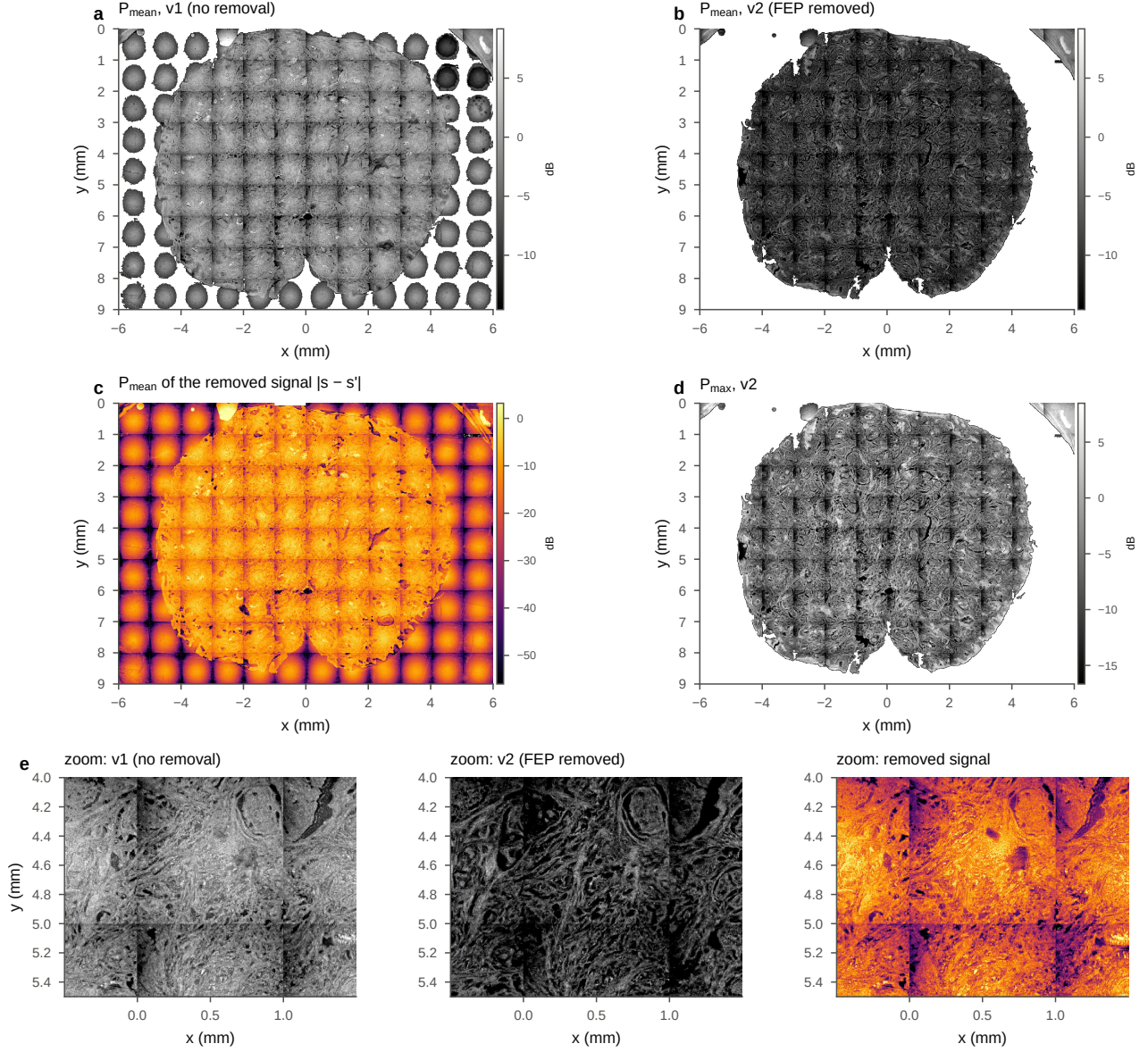


Fig. 5. En-face projections of the full volume and the removed signal. **a,b**, Tissue-only mean-amplitude projection P_{mean} (mean over the segmented tissue slab of each column; white/empty = no tissue) of the full 12×9 mm volume without (v1) and with (v2) film-reflection removal, on the same grey scale (4×4 block-averaged for display). Without removal the film reflections in tiles without tissue are segmented as tissue (round blobs) and brighten the projection inside the section. **c**, P_{mean} of the removed-signal volume $|s - s'|$, stitched with the same weights. **d**, Tissue-only maximum projection P_{max} of the v2 volume. **e**, Full-resolution zoom (2×1.5 mm across a tile corner) of **a-c**.

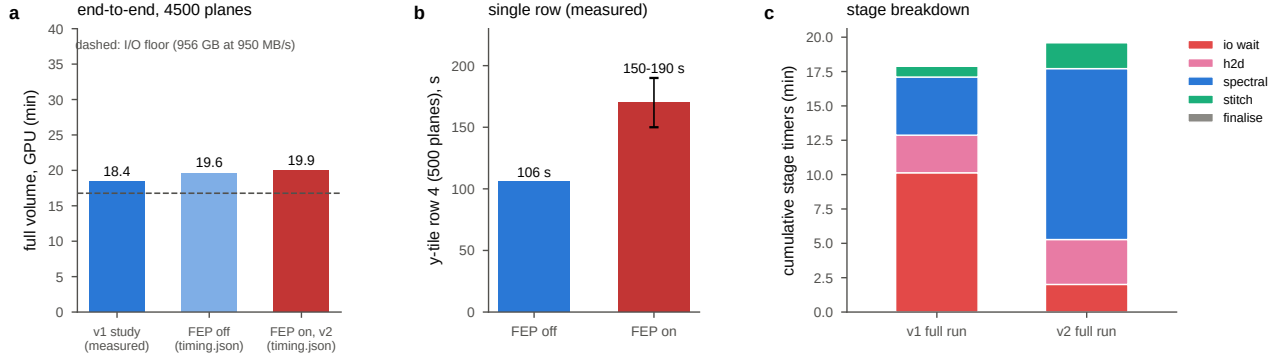


Fig. 6. Runtime. **a**, Full-volume GPU reconstruction time (4500 planes, end to end) of version 1 (measured in the original study) and of the current code with film removal off and on (`timing.json` of the full-volume run). Dashed, I/O floor. **b**, Time for one row of tiles (500 planes, 96 tiles) with removal off and on (range of repeated runs). **c**, Cumulative stage timers of the full runs (I/O wait, host-to-device copy, spectral processing, film removal, stitching, finalisation).

volume (same day, same disk), the reconstruction took 19.9 min with removal on (0.266 s per plane) versus 19.6 min with removal off, an overhead of only 2%. The full run is limited by reading the raw data from the external SSD: with removal off the pipeline waited 350 s for data, with removal on only 121 s, i.e. the additional GPU work is hidden behind the I/O (Fig. 6c). Start-up including learning the basis took 12 s, and the tissue mask, projections and removed-signal volume another 2.0 min. The complete reconstruction remains 77 times faster than the reference implementation without removal.

Discussion

Two practical obstacles of large-area spectral-domain OCT were addressed in one open pipeline. The reconstruction of an 864-tile, 956-GB volume now takes minutes rather than a day on a laptop, and the result is the established reference reconstruction, not an approximation: the GPU output differs from it by at most one quantisation step and can reproduce it bit-exactly. Because the GPU is idle for more than half of the run, the remaining lever is storage bandwidth; the same code on faster storage, or with raw data compressed losslessly, would approach its compute limit of under 0.1 s per plane.

The film-reflection removal exploits the one property that separates a specular interface from tissue in a single A-line: its complex depth profile is a shifted copy of the PSF. This makes the method local (it needs no neighbouring B-scans, so it is insensitive to the random B-scan-to-B-scan phase that defeated lateral averaging), and it keeps the signal outside a 17-sample window untouched. Learning the subspace from the scan absorbs residual dispersion and the actual spectral shape, which the theoretical PSF does not capture. The shrinkage to the background level is what makes the output visually and statistically plausible: a rule that is optimal for estimating the reflection (Wiener) is not optimal for estimating what lies beneath it, because the speckle component along the PSF subspace is removed as well. Keeping a fraction κ of the expected background energy is a simple, transparent compromise; the removed-signal volume documents exactly what was taken away.

The method has limitations. Tissue that is itself specular-like at the film interface—a flat, sharp tissue boundary coinciding with the film—cannot be separated from the film in a single A-line and is attenuated with it. Within ± 1 –2 samples of the surface, the image keeps the background intensity level but its axial speckle there is not the original. The decision thresholds were tuned on this dataset and may need adjustment for other films, objectives or spectrometers, although the basis itself is learnt automatically. Finally, other tile-periodic effects, such as vignetting and the tile seams of the stitching, are unrelated to the film and remain; they could be treated by flat-field estimation from the en-face projections that the pipeline now exports.

Methods

Imaging system and dataset

Data were acquired with a Thorlabs GAN632 spectral-domain OCT system (OCITY software) with an Olympus 20 $\times$ objective. [Sample description and preparation.] The tissue was mounted between FEP films. The tiled scan comprised 12 (x) $\times$ 9 (y) lateral tiles of 1 $\times$ 1 mm at 8 focal depths (864 tiles). Each tile holds 500 B-scans; each B-scan file contains 525 A-lines (25 apodisation A-lines and 500 image A-lines) of $N = 2048$ spectral pixels stored as int16, 956 GB in total. The output volume is 4500 (y) $\times$ 35 (z) $\times$ 6000 (x) voxels at 2 μ m isotropic spacing, covering $z \in [-29.5, 38.5]$ μ m relative to the tissue interface. All parameters are listed in Table 2; they are read from the scan metadata (`ScanInfo.json`, `Header.xml`, chirp file) or from the probe

preset, except where stated.

Spectral-domain signal model and per-A-line processing

For a lateral position $\mathbf{x} = (x, y)$ the spectrometer records, for pixel $j = 0, \dots, N - 1$,

$$I_j(\mathbf{x}) = S(k_j) \left[R_r + \sum_n R_n + 2 \sum_n \sqrt{R_r R_n} \cos(2k_j n_g z_n + \phi_n(k_j)) + \dots \right], \quad (1)$$

where S is the source spectrum, R_r and R_n the reference and sample reflectivities, z_n the depth of scatterer n , n_g the group index and ϕ_n a dispersion phase; the dots denote mutual interference terms of the sample^{2,4}. The wavelength of pixel j follows from the chirp c_j of the spectrometer calibration, $\lambda_j = [c_j(\lambda_{\max}^{-1} - \lambda_{\min}^{-1})/(N - 1) + \lambda_{\min}^{-1}]^{-1}$, with $[\lambda_{\min}, \lambda_{\max}] = [796, 1010]$ nm, and $k_j = 2\pi/\lambda_j$.

Apodisation and background subtraction. The mean of the $A = 25$ apodisation A-lines of the same B-scan file (recorded with the beam outside the sample) is subtracted from every image A-line, $I'_j(\mathbf{x}) = I_j(\mathbf{x}) - \frac{1}{A} \sum_{a=1}^A I_j(a)$.

k-linearisation. The spectrum is resampled onto a uniform wavenumber grid $\tilde{k}_i$, $i = 0, \dots, N - 1$, running linearly from max k to min k . With fractional source positions $n_j = (k_j - \min k)/(\max k - \min k)(N - 1)$ and target positions $q_i = N - 1 - i$, the sinc5 interpolation is

$$\tilde{I}_i = \sum_{j: |q_i - n_j| < 5} \text{sinc}(q_i - n_j) (I'_j - \bar{I}') + \bar{I}', \quad (2)$$

where $\bar{I}'$ is the spectral mean of the A-line. The weights depend only on the chirp; they are precomputed as a banded operator with $T \leq 14$ non-zero taps per output sample.

Window, dispersion compensation and Fourier transform. The resampled spectrum is multiplied by the complex window

$$W_i = \frac{h_i}{\text{rms}(h)} \exp(-i\beta(\tilde{k}_i - \langle \tilde{k} \rangle)^2), \quad (3)$$

with h a Hann window of length N , $\tilde{k}$ in rad nm^{-1} and the quadratic dispersion coefficient $\beta = 8.949 \times 10^7 \text{ nm}^2 \text{ rad}^{-1}$, which corrects the dominant second-order dispersion mismatch between the arms¹¹. The complex A-line is

$$s_m(\mathbf{x}) = \frac{1}{N} \sum_{i=0}^{N-1} W_i \tilde{I}_i(\mathbf{x}) e^{2\pi i i m / N}, \quad m = 0, \dots, N/2 - 1, \quad (4)$$

sampled at depths $z_m = m \Delta z$ with $\Delta z \approx \bar{\lambda}^2/(2n \Delta \lambda) = 1.434 \mu\text{m}$ in tissue ($n = 1.33$; $\bar{\lambda}$ and $\Delta \lambda$ the mean and span of the wavelengths). The image magnitude is $A_m(\mathbf{x}) = |s_m(\mathbf{x})|$; in version 2 the film removal (below) is applied to s_m first.

Optical-path (field-curvature) correction. The scan geometry adds a path-length offset $P(x, y) = p_1 x + p_2 y + p_3 x^2 + p_4 y^2 + p_5 xy$ (μm ; tile-local x, y in μm ; coefficients of the probe in Table 2). Nearest-neighbour resampling at $z_m + P$ on the uniform depth grid is an integer row shift $\delta(\mathbf{x}) = \lfloor P(\mathbf{x})/\Delta z + \frac{1}{2} \rfloor$,

$$A_m^c(\mathbf{x}) = A_{m+\delta(\mathbf{x})}(\mathbf{x}), \quad v_m(\mathbf{x}) = [0 \leq m + P(\mathbf{x})/\Delta z \leq N/2 - 1], \quad (5)$$

with invalid samples ($v = 0$) set to zero. Across a tile δ spans up to 37 rows ($\approx 53 \mu\text{m}$).

Focus weighting and stitching. Each tile is placed in the output frame at its lateral centre and focal depth, with its depth axis re-referenced so that $z = 0$ lies at the focus pixel $m_f = 433$. A focus weight

$$F_m = \exp\left(-\frac{(m - m_f)^2}{(2\sigma)^2}\right) + e^{-4.5}, \quad \sigma = 10 \text{ px}, \quad (6)$$

favours in-focus data (effective Gaussian standard deviation $\sqrt{2}\sigma$; the constant floor keeps out-of-focus data where no other tile contributes). Linear interpolation from tile coordinates to the output grid is separable; for

tile t it is written with an output-depth matrix $\mathbf{W}_z^{(t)}$ and an output-column matrix $\mathbf{W}_x^{(t)}$ (zero outside the tile). For every output plane y ,

$$\text{num}(z, x) = \sum_t \left[\mathbf{W}_z^{(t)} (\mathbf{A}^c \odot \mathbf{v} \odot \mathbf{F})^{(t)} \mathbf{W}_x^{(t)\top} \right]_{zx}, \quad \text{den}(z, x) = \sum_t \left[\mathbf{W}_z^{(t)} (\mathbf{v} \odot \mathbf{F})^{(t)} \mathbf{W}_x^{(t)\top} \right]_{zx}, \quad (7)$$

summing over the 96 tiles that contribute to the plane. Voxels with $\text{den} < e^{-4.5}(1 - 8\epsilon)$ (ϵ , machine precision of the accumulator; the factor protects voxels that sit exactly on the threshold) are set to NaN, and the output is

$$I(x, y, z) = 20 \log_{10}(\text{num}(z, x) / \text{den}(z, x)). \quad (8)$$

Quantisation. The volume is written as a uint16 BigTIFF with the global range $[c_1, c_2] = [\min I, \max I]$, $Q = \text{round}((I - c_1)/(c_2 - c_1) \cdot 65534) + 1$ and $Q = 0$ for NaN; 1 LSB = $(c_2 - c_1)/65534 \approx 1.02 \times 10^{-3}$ dB. The reference implementation quantises each plane to its own range first and re-quantises to the global range; this two-step rounding is available as an option and was used for the bit-exact comparison. Metadata are stored as JSON in the first page and as a sidecar file.

Model of the film reflection

Around a film surface at (fractional) depth $\mu(\mathbf{x})$ the complex A-line is

$$s_m(\mathbf{x}) = a(\mathbf{x}) e^{i\varphi(\mathbf{x})} p(m - \mu(\mathbf{x})) + t_m(\mathbf{x}) + \eta_m(\mathbf{x}), \quad (9)$$

with p the complex system PSF, a and φ the amplitude and phase of the reflection, t tissue speckle and η noise. For an ideal reflector and perfect dispersion compensation, Eq. (4) gives $p(m - \mu) = \frac{1}{N} \sum_i |W_i| e^{2\pi i i(m - \mu)/N}$. In a window of $2R + 1$ samples centred on the nearest integer m_0 of μ , the profiles for offsets $d = \mu - m_0 \in [-\frac{1}{2}, \frac{1}{2}]$ form a smooth one-parameter family, which is well approximated by a subspace of dimension $r \ll 2R + 1$ (Fig. 3c). Tissue speckle¹², by contrast, is a random superposition of PSFs at all depths and distributes its energy over the whole window; its expected energy within an r -dimensional subspace is about $r P_{\text{bg}}$ for a local power P_{bg} per sample (more for axially correlated speckle; Fig. 3d). The measured film amplitude varies by about $\pm 20\%$ between neighbouring A-lines and φ is random between B-scans, so the reflection cannot be predicted from neighbours; it can only be estimated within each A-line.

PSF basis

Theoretical basis. For 41 offsets d uniformly spaced in $[-\frac{1}{2}, \frac{1}{2}]$ we compute

$$a_d(m) = \text{IFFT}_n \{ |W_n| e^{-2\pi i n(m_0 + d)/N} \}(m), \quad m_0 = N/4, \quad (10)$$

take the samples $m_0 - R, \dots, m_0 + R$, normalise each to unit norm, and keep the first r left singular vectors of the $(2R + 1) \times 41$ matrix as $\mathbf{U}_{\text{th}}$.

Learnt basis. Before processing, 12 tiles are selected on a $4(x) \times 3(y)$ grid at the focal depth closest to $z = 0$, and five B-scans near the centre of each tile (every 4th of 20) are reconstructed to complex A-lines. Surfaces are detected as described below; for each surface and A-line the peak is located, the $(2R + 1)$ -sample segment $\mathbf{s}_k$ around it extracted and kept if it is clearly specular ($\|\mathbf{U}_{\text{th}}^H \mathbf{s}_k\|^2 / \|\mathbf{s}_k\|^2 \geq 0.75$) and at least as bright as the median segment. The kept segments are phase-aligned and normalised, $\hat{\mathbf{s}}_k = \mathbf{s}_k e^{-i \arg s_{k,R}} / \|\mathbf{s}_k\|$, at most 6000 are drawn at random, and the basis $\mathbf{U}$ is formed by the first r left singular vectors of $[\hat{\mathbf{s}}_1, \dots, \hat{\mathbf{s}}_K]$. If fewer than 50 segments are found, $\mathbf{U}_{\text{th}}$ is used. A basis can also be supplied as a file.

Film-reflection removal

The removal operates per tile on each batch of complex B-scans, before the magnitude, and only on the depth band $[r_0, r_1]$ that reaches the output after curvature correction (extended by $R + S$ samples).

- (1) *Surface detection.* The curvature-flattened magnitude $|s_{r+\delta(\mathbf{x})}(\mathbf{x})|$ is averaged over the B-scans of the batch and the central 70% of the A-lines, giving a profile $L(r)$ in dB. A row r is a surface candidate if $L(r) = \max_{|r' - r| \leq 3} L(r')$ and $L(r) - \text{med}_{51} L(r) \geq 6$ dB, where med_{51} is a 51-row running median (the local baseline). Candidates are accepted strongest first if they are more than R rows apart, up to four surfaces (top and bottom film surfaces and their back sides).
- (2) *Per-A-line position.* For surface r the expected native row is $r + \delta(\mathbf{x})$; the peak $\hat{m}(\mathbf{x}) = \arg \max_{|m - r - \delta(\mathbf{x})| \leq S} |s_m(\mathbf{x})|$ ($S = 4$) is smoothed with a 15-A-line lateral median and rounded, $\hat{m}(\mathbf{x})$, so that a bright speckle grain cannot capture it.

- (3) *Projection.* With $\mathbf{s}(\mathbf{x}) = (s_{\tilde{m}-R}, \dots, s_{\tilde{m}+R})^\top$, $\mathbf{c} = \mathbf{U}^\top \mathbf{s}$ and the specular-likeness $f = \|\mathbf{c}\|^2 / \|\mathbf{s}\|^2$.
- (4) *Soft decision.* $w = \text{clip}((\text{med}_{15}f - f_{\text{lo}})/(f_{\text{hi}} - f_{\text{lo}}), 0, 1)$, $f_{\text{lo}} = 0.35$, $f_{\text{hi}} = 0.60$, with med_{15} the lateral median over 15 A-lines.
- (5) *Shrinkage to the background level.* With the local background power

$$P_{\text{bg}} = \frac{1}{2B} \sum_{b=1}^B (|s_{\tilde{m}-R-b}|^2 + |s_{\tilde{m}+R+b}|^2), \quad B = 10, \quad (11)$$

the gain and the corrected segment are

$$g = \max\left(0, 1 - \sqrt{\frac{\kappa r P_{\text{bg}}}{\|\mathbf{c}\|^2}}\right), \quad \mathbf{s}' = \mathbf{s} - w g \mathbf{U} \mathbf{c}, \quad \kappa = 0.7. \quad (12)$$

For $g > 0$ the energy left along the subspace is $\|(1-g)\mathbf{c}\|^2 = \kappa r P_{\text{bg}}$, i.e. a fraction κ of what the background is expected to have there, independently of the film amplitude (a shrinkage in the spirit of soft thresholding¹³, applied to the subspace energy). The Wiener-type rule $g_W = \max(0, 1 - r P_{\text{bg}} / \|\mathbf{c}\|^2)$ leaves $(r P_{\text{bg}})^2 / \|\mathbf{c}\|^2$, which vanishes for strong reflections and produced a dark band on the data (film-only tile: -23.0 dB at the surface against a -16.0 dB background); $\kappa = 0$ subtracts the full projection.

Samples outside $\tilde{m} \pm R$ are bit-identical to the input; inside the window only the r components along $\mathbf{U}$ change. The heavy operations (gathers, projections, updates) run on the device in the batch; only $(B \times n_x)$ arrays of peak positions and f are transferred for the median filters.

Removed-signal volume. The subtracted component $\Delta \mathbf{s} = \mathbf{s} - \mathbf{s}' = w g \mathbf{U} \mathbf{c}$ (summed over surfaces) is kept as a complex array; its magnitude $|\Delta s_m|$ is curvature-corrected, weighted and stitched with exactly the same $\mathbf{v}$, $\mathbf{F}$, $\mathbf{W}_z$, $\mathbf{W}_x$ and denominator as the image, and written as a second volume $I_{\text{rem}} = 20 \log_{10}(\text{num}_{\text{rem}}/\text{den})$.

Tissue segmentation and tissue-only en-face projections

The stitched dB volume $I(x, y, z)$ is processed in chunks of 256 y -planes directly from the memory-mapped result (three streaming passes), so the step works for volumes larger than the main memory. (i) *Lateral smoothing.* Per depth plane the amplitude $A = 10^{I/20}$ is averaged over an $L \times L$ window in (x, y) ($L = 30 \mu\text{m} = 15$ pixels) by normalised convolution, ignoring voxels without data:

$$S(x, y, z) = 20 \log_{10} \frac{(\mathbb{K}_L * A \chi)(x, y, z)}{(\mathbb{K}_L * \chi)(x, y, z)}, \quad (13)$$

with χ the indicator of finite voxels and $\mathbb{K}_L$ the box kernel. Averaging over the speckle separates the volume-scattering tissue from the noise floor. (ii) *Threshold.* The threshold τ is Otsu's threshold of the histogram of all S values (0.05 dB bins), i.e. the value maximising the between-class variance $\omega_0(\tau) \omega_1(\tau) [\mu_0(\tau) - \mu_1(\tau)]^2$; the voxel mask is $M = S > \tau$. (iii) *Footprint.* A column is tissue if it contains at least two masked voxels; connected components smaller than 0.005 mm^2 are removed, and enclosed non-tissue regions smaller than 0.5 mm^2 that do not touch the border of the field (dark lumens, the vignettied tile corners) are added, giving the footprint Ω . (iv) *Slab.* Per column the first and last masked depths $z_{\text{top}}(x, y)$, $z_{\text{bot}}(x, y)$ are smoothed laterally over $50 \mu\text{m}$ (normalised convolution over the measured columns) and interpolated into the filled holes (nearest measured value); the slab is $Z(x, y) = \{z : \lfloor z_{\text{top}} \rfloor \leq z \leq \lceil z_{\text{bot}} \rceil, I(x, y, z) \text{ finite}\}$ for $(x, y) \in \Omega$ and empty otherwise. Dark structures between the two surfaces are therefore part of the tissue. (v) *Projections.*

$$P_{\text{mean}}(x, y) = 20 \log_{10} \left(\frac{1}{|Z(x, y)|} \sum_{z \in Z(x, y)} 10^{I(x, y, z)/20} \right), \quad P_{\text{max}}(x, y) = \max_{z \in Z(x, y)} I(x, y, z), \quad (14)$$

and both are undefined (NaN) for $Z(x, y) = \emptyset$; P_{mean} averages amplitudes, as the stitching does, not dB values. Both projections and the slab-thickness map $|Z| \Delta z$ are saved as calibrated 2D TIFF images (ImageJ, μm per pixel) with PNG previews. For the removed-signal volume the projection is taken over all finite depths. The projection can be recomputed from a kept float volume with other settings without reconstructing again. To quantify tile-periodic artefacts we fold $10^{P_{\text{mean}}/20}$ onto one 500×500 tile by averaging over all tiles, smooth it with a Gaussian ($\sigma = 5 \text{ px}$) and report the difference between its 95th and 5th percentiles in dB ('tile-periodic modulation').

GPU implementation

The code is written against an array-module abstraction (NumPy¹⁴ or CuPy¹⁵; PyTorch for Apple GPUs), so the same routines run on CPU and GPU. Raw files of one row of tiles (96 tiles $\times$ 500 B-scans) are read tile-major with unbuffered `readinto` into preallocated int16 buffers by a pool of 8 threads, with up to 16 batches of 50 B-scans in flight (reduced automatically if free memory is short), so reading overlaps with computation. Each batch (25,000 A-lines) is copied to the device and processed by one fused CUDA kernel (one thread per output spectral sample) that subtracts the apodisation and spectral means, applies the banded sinc5 operator, re-adds the mean and multiplies by the complex window, writing complex64 spectra; a batched cuFFT computes Eq. (4). The curvature correction is a single gather with the precomputed shift map, restricted to the rows that can reach the output, and stitching is two batched GEMMs (cuBLAS) for numerator and denominator into accumulators for the whole row of tiles (500 $\times$ 35 $\times$ 6000 $\times$ 2 float32, $\approx$ 840 MB) that remain in GPU memory. After the last tile of a row the NaN mask and the dB conversion are applied on the device and one array is copied back. On the CPU the fused step is a Numba kernel parallelised over A-lines and the FFT is multithreaded pocketfft^{16,17}. Single precision is used throughout; its error relative to double precision ($\approx 10^{-4}$ dB) is ten times below 1 LSB. Timers synchronise the device and are exclusive. Benchmarks used an NVIDIA RTX 3080 Ti Laptop GPU (16 GB), an AMD Ryzen 9 6900HX (8 cores, 16 threads), 14 GB RAM and an external USB SSD (≈ 950 MB s⁻¹ sequential read).

Validation

Each stage was compared with the unmodified reference functions run on the same raw B-scan in MATLAB R2024a (maximum relative errors 2.4×10^{-12} for the resampled spectrum, 8.5×10^{-13} for the complex scan in double precision; zero mismatches in the curvature validity map), and the complete reconstruction with the reference volume voxel by voxel. For the film removal, synthetic ground truth was generated by adding PSF reflections (Hann window, sub-pixel tilt $0.8 \sin(2\pi x/n_x)$ samples, random phase per A-line, $\pm 20\%$ uniform amplitude jitter) to complex Gaussian speckle filtered axially with a 5-tap Hann kernel and to a 30-dB weaker noise floor (4 $\times$ 200 A-lines). Errors are energies in the 17-sample window relative to the tissue. The same generators are used in the unit tests of the software, which require a residual below -6 dB for reflections 10–30 dB above tissue, an empty-background level within ± 3 dB and a change below -20 dB for tissue without reflection. For real data, single B-scans of a film-only tile and a tissue tile were processed on the CPU with the production code, and a full row of tiles was reconstructed on the GPU with removal off and on. The figures and all numbers in this paper are generated by a script from these results.

Data availability

The raw tiled dataset (956 GB) and the reconstructed volumes are available from the corresponding author on reasonable request [repository and accession to be added]. The intermediate results used for the figures (single-tile complex B-scans, the reconstructed row of tiles with and without removal, en-face projections and timing records) are provided with the code.

Code availability

Version 1 of the reconstruction software (GPU/CPU port of the myOCT tiled-scan reconstruction, web application, validation scripts) is available at https://github.com/atesfet/oct_recon_AF_v1. Version 2 (film-reflection removal, en-face projections, removed-signal volume) will be available at [repository URL for v2]. The script that generates all figures and numbers of this paper is included.

Acknowledgements

[Acknowledgements and funding.]

Author contributions

[Author contributions.]

Competing interests

[Competing interests statement.]

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Table 2. Reconstruction and film-removal parameters for the dataset (configuration names of the software in typewriter font).

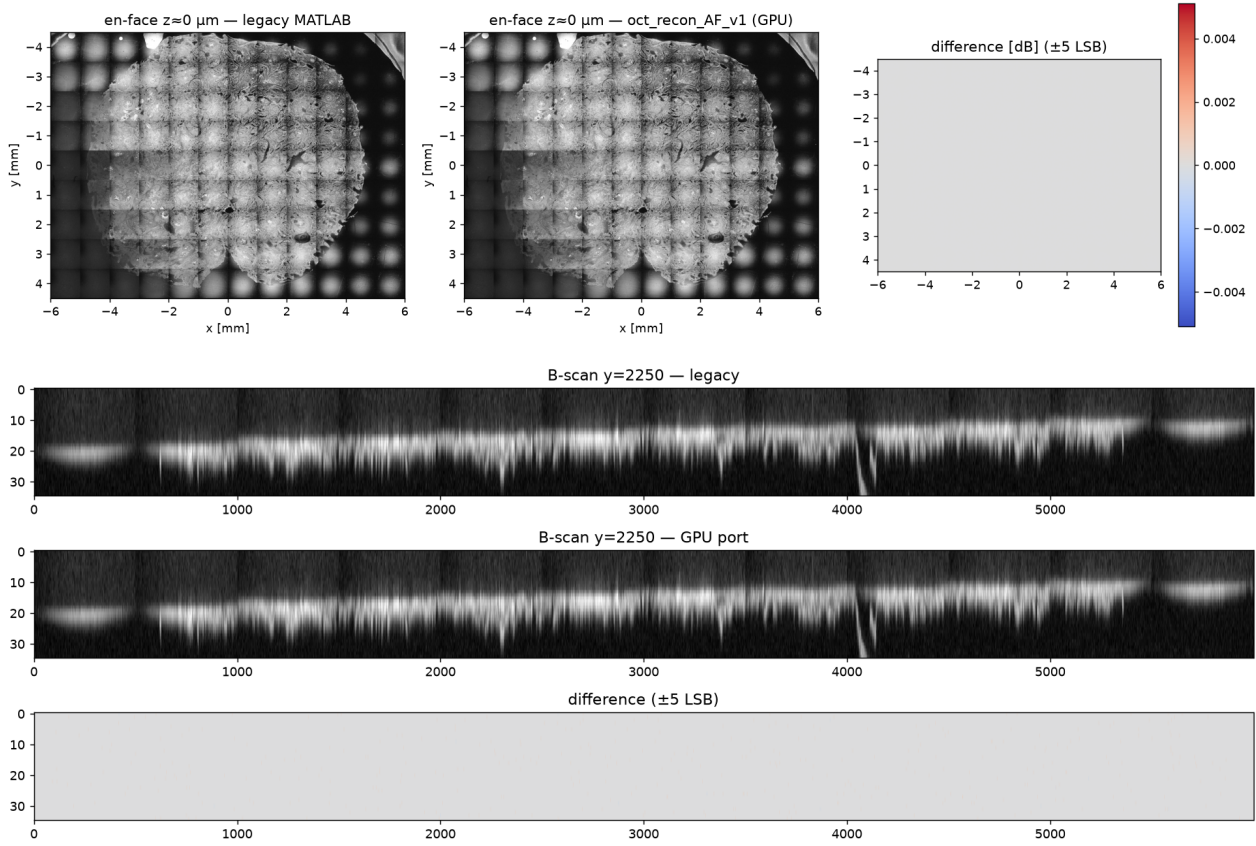
Parameter	Value	Source / meaning
Spectral pixels N , A-lines, apodisation lines	2048, 500, 25	Header.xml
Wavelength range	796–1010 nm	system preset (GAN632)
k-linearisation	sinc5 (interp_method)	reference default
Dispersion β	$8.949 \times 10^7 \text{ nm}^2 \text{ rad}^{-1}$	probe preset; reproduces reference bit-exactly
Refractive index n	1.33	ScanInfo.json
Depth pixel Δz	1.43385 μm (tile), 2 μm (output)	derived; output = lateral pixel size
Optical-path polynomial $p_{1..5}$	0.0099168, 0.010817, -0.0001907 , -0.00026705 , 4.8047×10^{-12}	probe (ScanInfo.json)
Focus pixel m_f , width σ	433 px, 10 px	focus-calibration file; objective preset
Output depth range	-30 to $+40 \mu\text{m}$ (crop_z_range_mm)	range of focal depths
Batch, reader threads, prefetch	50 B-scans, 8, 16 batches	execution
Precision	float32 / complex64	execution
FEP removal	on (fep_removal)	off = version-1 output
Basis rank r , half-window R	3, 8 (fep_rank , fep_half_window)	17-sample window
Peak search S , lateral median	± 4 , 15 A-lines (fep_search , fep_lateral_median)	
Weight ramp f_{lo} , f_{hi}	0.35, 0.60 (fep_frac_lo/hi)	specular-likeness
Detection threshold, max. surfaces	6 dB, 4 (fep_detect_min_db , fep_max_surfaces)	above 51-row median
Background rows B , level κ	10, 0.7 (fep_keep_level)	$\kappa = 0$: full projection
Basis	learnt, 12 tiles, ≤ 6000 segments (fep_learn_basis)	fallback: theoretical
xy projections, removed volume	on (version 2)	off = version-1 output

Fourier domain optical coherence tomography. *J. Biomed. Opt.* **7**, 457–463 (2002).

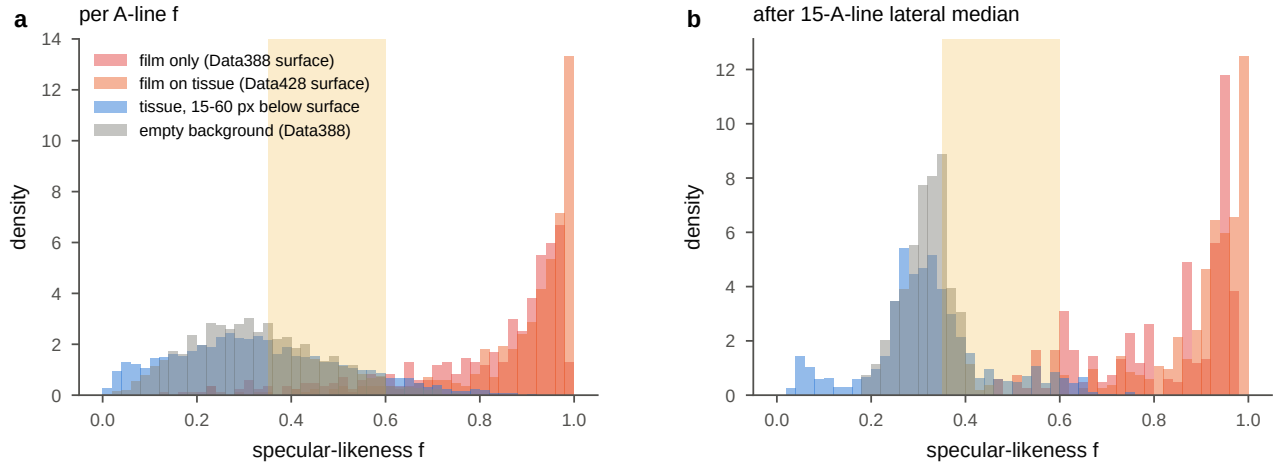
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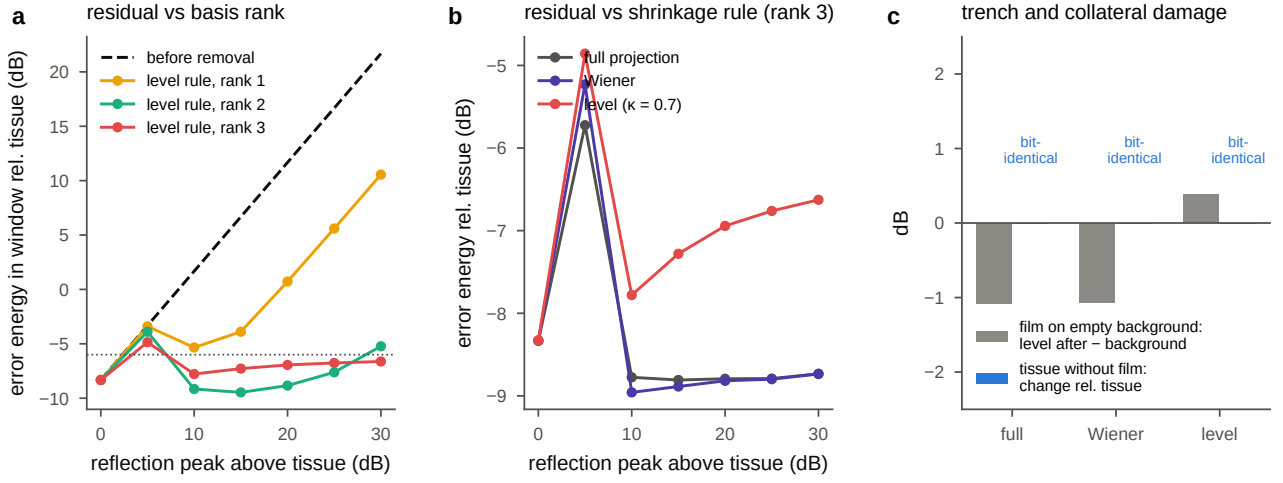
Extended Data



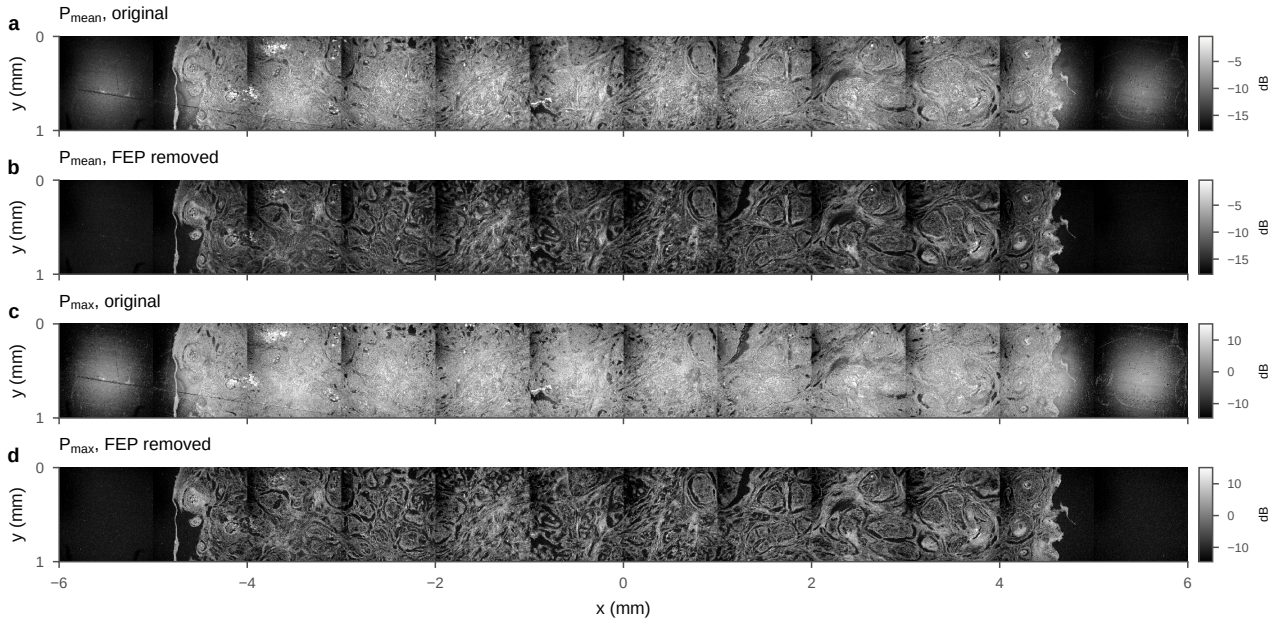
Extended Data Fig. 1. Agreement with the reference reconstruction (version 1, film removal off). Top: en-face plane at $z \approx 0$ over the full 12×9 mm field from the reference MATLAB reconstruction, from the GPU pipeline, and their difference (colour range ± 5 LSB). Bottom: output B-scan (plane 2250) from both and their difference. Over all 945,000,000 voxels the maximum difference is 1 LSB (1.02×10^{-3} dB), 90.7 % of the voxels have identical codes and the masks of missing data are identical; the remaining 1-LSB differences come from the reference's two-step quantisation.



Extended Data Fig. 2. Decision statistic on real data. Specular-likeness $f = \|U^H s\|^2 / \|s\|^2$ with the learnt rank-3 basis for segments centred on the film surface of a film-only tile and of a tissue tile, for tissue 15–60 samples below the surface and for empty background (one B-scan each, A-lines 40–460). **a**, Per A-line. **b**, After the 15-A-line lateral median used by the method. Yellow, ramp of the weight w . After the median, 95 % of the film positions receive full weight, while 66 % of the tissue and 75 % of the background positions receive zero weight; the remainder receive small weights, which is why processing is additionally restricted to detected surface rows.



Extended Data Fig. 3. Synthetic ground truth. PSF reflections (sub-pixel tilt, random phase, $\pm 20\%$ amplitude jitter) injected into simulated speckle tissue; error energy in the 17-sample window relative to the tissue. **a**, Level rule ($\kappa = 0.7$) with bases of rank 1–3 (theoretical PSF). Dashed, error without removal; dotted, -6 dB acceptance limit of the unit tests. Reflections up to ≈ 5 dB above the tissue fall below the specular-likeness ramp and are not removed. **b**, Rank 3 with the full projection, the Wiener rule and the level rule. The level rule deliberately keeps $\kappa r P_{bg}$ along the subspace, hence a residual about 2 dB higher in tissue. **c**, Reflection 25 dB above an empty noise background: level in the surface rows after removal minus the noise level (grey); tissue without any reflection processed at the same row: the output is bit-identical because the weight w is zero (blue).



Extended Data Fig. 4. En-face projections of one row of tiles (row 4, 12×1 mm, all 35 output depths) without and with film-reflection removal, computed from the stitched volumes with the definitions in Methods. **a,b**, P_{mean} ; **c,d**, P_{max} ; each pair on the same grey scale. Median P_{mean} -10.2 versus -14.5 dB; median P_{max} 1.8 versus -8.1 dB; tile-periodic modulation of P_{mean} 7.5 versus 4.5 dB.